Shreyas Gupta

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Summary

AI Researcher and Machine Learning Engineer pursuing Integrated M.Tech in Artificial Intelligence with hands-on expertise in generative AI, agentic AI systems, multi-agent architectures, RAG pipelines, transformer-based models, and graph AI. Proven ability to design novel AI algorithms, build production-grade AI systems, and collaborate across cross-functional teams. Winner of Agentic AI Hackathon 2025 (NielsenIQ & TCS). Passionate about advancing research in neural architectures, knowledge representation, and autonomous AI systems.

Education

SRM Institute of Science and Technology — Chennai, India
Integrated M.Tech in Artificial Intelligence (CGPA: 8.29/10) — *Aug. 2022 – May 2027*

- Coursework: Machine Learning, Deep Learning, Neural Network Architectures, Statistical Learning Theory, Natural Language Processing, Computer Vision, Graph Neural Networks, Reinforcement Learning, AI Agents & Multi-Agent Systems

Experience

AI/ML Engineer – Research Intern — May 2026 – July 2026
Revenuxe — *Bangalore, India*

- Designed and implemented RAG (Retrieval-Augmented Generation) pipelines and agentic AI architectures with multi-agent orchestration using Python, FastAPI, and LangChain across two enterprise AI products.
- Researched and developed novel prompt engineering strategies and attention-based retrieval mechanisms to improve generative AI model performance by 35% on domain-specific benchmarks.
- Architected autonomous AI agent systems with tool-use capabilities, chain-of-thought reasoning, and knowledge representation modules for complex enterprise workflows.
- Shipped production-grade full-stack AI applications, migrated inference backends to AWS (EC2, Lambda, S3), and Dockerized ML model deployments for scalable serving.

Cybersecurity Research Intern – AI & Graph Systems — Dec 2025 – March 2026
Indian Institute of Technology, Guwahati — *Guwahati, India*

- Conducted research on graph-based AI models for cybersecurity vulnerability detection, designing modular architecture with CLI, parser, validator, loader, query engine, and visualization modules.
- Implemented custom LadyBugDB embedded graph database with novel graph traversal algorithms for representing function call graphs and control flow edges – applicable to Graph AI and knowledge representation.
- Applied causal reasoning techniques to identify vulnerability propagation paths in software dependency graphs, contributing to explainable AI for security analysis.

Open Source Contributor – AI Engineer — Dec 2025 – Present
NgKore Foundation (5G Core) — *Remote*

- Built autonomous Python AI agents leveraging LLM-based reasoning and multi-agent collaboration to automate monitoring, anomaly detection, and configuration management for 5G core network deployments.
- Designed agent architectures with planning, tool-use, and self-correction capabilities – demonstrating practical implementation of agentic and multi-agentic systems.
- Contributed features and bug fixes via GitHub PRs, reviewed and merged by senior maintainers in open-source AI community.

Business Strategy & Data Analytics Intern — May 2024 – July 2024
To-Let Globe — *Lucknow, India*

- Applied statistical analysis and data-driven inference on 10,000+ rental listings using SQL and Python; built Power BI dashboards to visualize patterns and support data-driven decision-making.

Research Projects

BharatGreen AI | *Python, FastAPI, Next.js, TypeScript, NVIDIA NIM, Generative AI* — Jan 2026 – March 2026

- Designed and developed AI sustainability research platform integrating generative AI models (NVIDIA NIM) for energy forecasting, carbon footprint analysis, and water usage optimization across Indian regions.
- Implemented what-if analysis engine using causal inference and temporal modeling to simulate carbon-aware scheduling scenarios – advancing research in AI for social good and UN SDGs.
- Developed real-time multimodal data processing pipeline handling temporal sensor data, geospatial inputs, and text-based policy documents using transformer-based architectures.

Light Pollution Sentinel | *Python, Next.js, PostgreSQL, OpenAI, Docker, AI Agents* — July 2025 – Sept 2025

- Built autonomous AI agent pipeline for detecting light-pollution hotspots from NASA satellite data across 742 Indian districts using concept discovery and extraction techniques.
- Developed explainable AI insights module generating severity-based alerts with natural language explanations, demonstrating research in explainable AI and knowledge extraction.
- Shipped production system with real-time dashboard featuring heatmaps, AI-generated analysis, and automated email alerting – end-to-end AI research to deployment pipeline.

Awards & Achievements

Winner – Agentic AI Hackathon 2025 | NielsenIQ & TCS – Built ALPS (autonomous multi-agent system) Sep 2025
Member – Indian Society for Technical Education (ISTE) | SRM Institute Mar 2025 – Present

Technical Skills

AI/ML Frameworks: TensorFlow, PyTorch, LangChain, OpenAI API, Google Gemini API, NVIDIA NIM, Hugging Face Transformers, scikit-learn, NLTK, Seaborn, Streamlit
AI Research Areas: Generative AI (LLMs), Agentic AI, Multi-Agent Systems, RAG, Transformer Architectures, Attention Mechanisms, Graph AI, Explainable AI, Causal Reasoning, Knowledge Representation, Concept Discovery
Programming Languages: Python, Java, C/C++, SQL (PostgreSQL), JavaScript, TypeScript
Frameworks & Tools: FastAPI, React, Next.js, Node.js, Flask
Infrastructure & DevOps: AWS (EC2, Lambda, S3), Azure, Docker, Linux, Git, GitHub, CI/CD
Research Tools: Jupyter Notebook, Anaconda, VS Code, LaTeX, Weights & Biases

Certifications

TATA GenAI Powered Data Analytics, Forage (Oct '25) | Deloitte Technology Job Simulation, Forage (Jun '25) | Competitive Programming (Excellence), Coding Ninjas (Jun '24) | Programming in Java, NPTEL (Oct '23)